7.1 The nucleus

At the end of this topic, you should be able to:

- **identify particles inside the nucleus**
- **discus the historical origins of the particles of the nucleus**
- **define and discuss isotope of an atom**
- **describe the relationships between nuclear mass and binding energy of isotopes;**
- **calculate binding energy of the nucleus of different isotopes;**
- **identify the nuclear forces, (strong nuclear force, weak nuclear force)**

7.2 Radioactivity

By the end of this section you should be to:

- **define the terms radio-activity, and distinguish between the three types of emissions, alpha radiation, beta- radiation and gamma- radiation**
- **recognize and discuss dangers of ionizing radiation;**
- **discuss radioactive dating and radiation detectors;**
- **identify radioactive sources in the school, home, and workplace and**
- **recommend protective measures to the school community.**

7.3 Use of nuclear radiation

By the end of this section you should be able to:

- **investigate the benefits and hazards (nuclear waste) of nuclear energy production;**
- **discuss and appreciate the application of nuclear medicine**

7.4 Nuclear Reaction and Energy Production

By the end of this section you should be able to:

- **distinguish between the two types of nuclear reaction;**
- **appreciate the sun as a big nuclear reactor.**
- **discuss the use (energy production, production of important isotopes, etc.)**
- **and misuse (nuclear bomb) of artificial nuclear reaction.**

7.5 Safety Rules Against Hazards of Nuclear Radiation

By the end of this section you should be able to:

- **implement safety rules against hazards of nuclear radiation**